

Project Name : Go Gold

Group Members

- a. Wichada Chaiprasertsud 6532157921
- b. Chayodom Vongbunsin 6531311021
- c. Natouch Srikiatden 6531314921

Problem Description

Small and medium-sized gold shops still rely on paper or spreadsheets to manage pawn tickets, leading to errors in interest calculations, lack of traceability when re-issuing tickets, and poor customer engagement. Currently, training new staff to accurately calculate interest and manage pawn tickets takes significant time and effort, and it often takes weeks for them to become proficient. The system should simplify ticket issuance, interest renewal, redemption, top-up, and maintain a complete audit trail. It should also provide automated notifications, financial summaries, and customer support features like reservation requests for items.

Target Customer

Admin (Owner): Full access, approve transactions, manage staff, generate reports.

Staff: Create and manage pawn tickets, process renewals, redemptions.

Customer: View tickets, request item reservation, receive reminders.

Scenario (use-case & description)

Use Case Name: Issue Pawn Ticket	ID: UC-01	Important Level: High
Primary Actor: Staff (Admin can also perform)	Use Case Type: Business / Transactional	
Stakeholders and Interests: Staff: wants to create tickets quickly without errors Admin: wants traceability and approval Customer: wants accurate valuation and fair loan		
Brief Description: Staff issues a new pawn ticket with gold item details and loan amount suggestion.		
Trigger: Customer brings gold item to pawn. Type: External		
Relationships: Depends on Gold Price Service.		
Precondition: Customer is registered; Staff authenticated.		
Postcondition: Ticket created, awaiting approval (if Staff).		
Normal Flow of Events: - Staff inputs gold item details. - System fetches gold price, suggests loan. - Staff submits ticket. - Admin approves. - Ticket becomes active.		
Subflows: None.		
Alternate/Exceptional Flow: Admin directly creates → auto-approved.		

Use Case Name: Renew Interest	ID: UC-02	Important Level: High
Primary Actor: Staff (Admin can also perform)		Use Case Type: Business / Transactional
Stakeholders and Interests: Staff: wants to renew tickets quickly without errors Admin: ensures interest is collected properly. Customer: wants to keep pawn valid without losing item.		
Brief Description: Customer pays interest at the shop to extend the due date.		
Trigger: Due date approaching. Type: External		
Relationships: Relies on Interest Calculation module.		
Precondition: Active ticket exists.		
Postcondition: Due date extended; renewal recorded.		
Normal Flow of Events: - Customer comes to shop. - Staff opens ticket and calculates interest. - Staff confirms payment received in person. - Admin approves. - System updates due date. - Audit log recorded.		
Subflows: Batch renewals by Admin.		
Alternate/Exceptional Flow: Customer cannot pay full interest → renewal not processed.		

Use Case Name: Redeem Item	ID: UC-03	Important Level: High
Primary Actor: Staff (Admin can also perform)		Use Case Type: Business / Transactional
Stakeholders and Interests: Customer: wants to retrieve pawned gold. Staff: ensures correct settlement. Admin: maintains financial control.		
Brief Description: Customer redeems item by paying principal + interest.		
Trigger: Customer requests redemption at shop. Type: External		
Relationships: None (manual payment).		
Precondition: Active pawn ticket.		
Postcondition: Ticket marked Redeemed; item returned.		
Normal Flow of Events: - Customer comes to shop. - Staff checks owed amount (principal + interest) from the system. - Staff confirms payment received. - Admin approves. - Ticket updated to Redeemed. - Audit recorded.		
Subflows: Batch renewals by Admin.		
Alternate/Exceptional Flow: Customer cannot pay full interest → renewal not processed.		

Use Case Name: Adjust Loan (Top-up/Reduce Principal)	ID: UC-04	Important Level: Medium
Primary Actor: Staff (Admin can also perform)		Use Case Type: Business / Adjustment
Stakeholders and Interests: Customer: wants to receive more cash (top-up) or reduce debt (reduce principal). Staff: wants a clear process to adjust tickets without calculation errors. Admin: requires transparency and traceability of adjustments.		
Brief Description: When a customer requests a top-up or reduction of principal, the shop first clears the interest of the old ticket up to the current date. Then a new ticket (re-issue) is created with the updated principal amount.		
Trigger: Customer visits the shop and requests a loan adjustment. Type: External		
Relationships: Linked with Re-issue Ticket flow, Audit Log Service.		
Precondition: An active ticket exists.		
Postcondition: Old ticket closed, marked with cleared interest. New ticket issued under the same master id with updated principal.		
Normal Flow of Events: - Customer requests top-up or reduction. - Staff opens the old ticket. - System calculates interest up to the current date. - Staff records the payment of old interest. - Staff enters the new principal (higher or lower). - System issues a new ticket (ticket_id new, master_id unchanged). - If top-up → system calculates additional cash to give to customer and records the payout. - If reduce principal → system records the extra amount received from customer. - Staff confirms payment received. - Admin approves. - Ticket updated. - Audit Service logs the link between old and new ticket.		
Subflows: Admin can directly perform the adjustment → auto-approved.		
Alternate/Exceptional Flow: If the customer cannot pay the outstanding interest → adjustment request is denied.		

Use Case Name: Generate Monthly Summary Report	ID: UC-05	Important Level: Medium
Primary Actor: Staff (Admin can also perform)		Use Case Type: Business / Reporting
Stakeholders and Interests: Admin: requires financial summaries for decisions. Business: needs compliance and overview.		
Brief Description: Admin generates monthly summary report of transactions.		
Trigger: End of month.. Type: External		
Relationships: Collaborates with Ticket, Interest, and Audit Services.		
Precondition: Transactions recorded.		
Postcondition: Report generated (PDF/Excel).		
Normal Flow of Events: - Admin requests monthly report. - System aggregates data (issued, renewed, redeemed). - Generates report in chosen format. - Admin downloads.		
Subflows: Daily reports.		
Alternate/Exceptional Flow: System fails → retry or manual aggregation.		

Use Case Name: Notify Customer of Due Date	ID: UC-06	Important Level: Medium
Primary Actor: System (Notification Service)		Use Case Type: Automated / Reminder
Stakeholders and Interests: Customer: wants timely reminders.		
Brief Description: System sends reminders before due date.		
Trigger: Scheduled daily check. Type: System-triggered		
Relationships: Notification Service.		
Precondition: Ticket has due date, contact info available.		
Postcondition: Customer notified.		
Normal Flow of Events: - System checks due dates daily. - Identifies tickets expiring soon. - Sends SMS/Email reminder. - Audit logs notification.		
Subflows: None.		
Alternate/Exceptional Flow: Invalid contact info → notification fails.		

Use Case Name: Request to Hold Item	ID: UC-07	Important Level: Medium
Primary Actor: Customer		Use Case Type: Customer Service
Stakeholders and Interests: Customer: wants extra time to pay.		
Brief Description: Customer requests shop to hold pawned item beyond due date.		
Trigger: Customer foresees late renewal. Type: External		
Relationships: Notification Service alerts staff.		
Precondition: Ticket near expiry.		
Postcondition: Hold request logged and possibly approved.		
Normal Flow of Events: - Customer logs in and submits request. - System records request. - Notification sent to staff. - Admin/Staff approves/denies. - Status updated in system.		
Subflows: None.		
Alternate/Exceptional Flow: Request denied → system notifies customer.		

Functional Requirements

User Registration & Login System

- The system shall allow customers to register using their phone number as a unique identifier.
- The system shall allow customers to log in using OTP sent via SMS (with email fallback).
- The system shall allow admins to create and manage staff accounts.
- The system shall allow customers, staff, and admins to log out securely.

Pawn Ticket Management System

- The system shall allow staff to issue new pawn tickets with item details (type, weight, purity).
- The system shall allow admins to approve or reject tickets created by staff.
- The system shall allow staff or admins to re-issue a ticket with a new ticket ID, linked to the old one via master_id.
- The system shall allow staff to upload images of pawned items to attach to tickets.
- The system shall allow admins to edit or batch-update multiple tickets.

Interest & Loan Adjustment System

- The system shall calculate interest automatically based on default and per-ticket formulas.
- The system shall allow staff to record renewal of interest **and require admin approval** before it is finalized.
- The system shall allow staff to process redemption requests **but require admin approval** to close the ticket.
- The system shall allow staff/admin to request principal adjustments (Top-up/Reduce) **but require admin approval** to issue the new ticket.

Customer Portal System

- The system shall allow customers to view their own active and historical pawn tickets.
- The system shall allow customers to request an item hold (extension to keep item after due date).
- The system shall notify customers of ticket due dates and request results.

Reporting System

- The system shall allow admins to generate daily financial summaries.
- The system shall allow admins to generate monthly financial reports in PDF and Excel format.
- The system shall aggregate data on issued tickets, renewals, redemptions, and outstanding balances.

Notification System

- The system shall notify customers by SMS/email when their ticket is near its due date.
- The system shall notify admins when a customer submits a hold request.
- The system shall notify staff and admins of tickets requiring approval.

Non-Functional Requirements

Operational

- The system shall display properly on both desktop and mobile devices.
- The system shall support major browsers and OS (Windows, MacOS, iOS, Android).
- The system shall back up all ticket and transaction data daily.
- The system shall retain audit logs for at least 3 years.

Performance

- The system shall respond to ticket or interest calculation requests within 1 second.
- The system shall operate correctly with an uptime of at least 99.5%.

Security

- The system shall enforce OTP + JWT authentication for customers.
- The system shall provide role-based access control (Admin, Staff, Customer).
- The system shall store all audit logs as append-only and tamper-proof.
- The system shall protect personal and financial information according to PDPA.

Cultural and Legal

- The system shall provide a Thai language user interface, with English as optional.
- The system shall use Thai Baht (THB) as the default currency.
- The system shall comply with Thailand's Computer Crime Act B.E. 2560 (2017) and Personal Data Protection Act B.E. 2562 (2019).

Usability

- The system shall be designed with a responsive UI.
- The system shall provide intuitive navigation for both staff and customers.
- The system shall minimize manual calculation steps to reduce staff workload.

Architecture Decision Records (ADRs)

ADR-001: Frontend Framework

Context

The system requires a modern frontend framework to deliver a responsive and SEO-friendly UI. Options considered: React with CRA, Vue, Angular, and Next.js. Next.js provides server-side rendering, routing, and SEO improvements out-of-the-box, while React CRA lacks built-in SSR and Angular is too heavy for the project scope.

Decision

Use Next.js (React) as the frontend framework.

Status

Accepted

Consequences

- Gains performance and SEO benefits from SSR.
- Developers can reuse React ecosystem libraries.
- Requires some learning curve for SSR/ISR concepts.

ADR-002: CSS/UI Framework

Context

The UI must be responsive and customizable. Options: TailwindCSS, Ant Design, Bootstrap. Ant Design provides ready components but can be rigid. Bootstrap is outdated in style. TailwindCSS offers utility-first design, responsive flexibility, and modern styling.

Decision

Adopt TailwindCSS for UI styling.

Status

Accepted

Consequences

- Enables fast prototyping with utility classes.
- Provides responsive and mobile-first support.
- Requires consistent coding practices to avoid messy HTML.
- Lacks prebuilt enterprise UI components; custom design work needed.

ADR-003: Backend Language & Framework

Context

The backend must handle performance-sensitive operations, modular business logic, and future scalability. Options: Node.js (NestJS), Django, Go (Golang). Go provides high performance, simplicity, and strong concurrency support. Team prefers a statically typed language for backend reliability.

Decision

Use Go (Golang) for backend development, building REST APIs and potentially gRPC in future.

Status

Accepted

Consequences

- Higher performance for concurrent workloads compared to Node.js.
- Requires developers to learn Go's idioms and patterns.
- Some web libraries are less mature than in Node.js ecosystem.

ADR-004: Authentication & OTP Strategy

Context

Most customers use mobile numbers; emails are less reliable. Options: SMS OTP, Email OTP, App-based OTP. SMS OTP is most convenient.

Decision

Use SMS OTP as primary authentication with email fallback, and issue JWT sessions post-login.

Status

Accepted

Consequences

- Easy customer access.
- Needs SMS gateway integration and cost handling.
- Requires anti-abuse measures (rate limit, OTP expiry).

ADR-005: Notification System

Context

The system must notify customers of due dates, approvals, and hold requests. Options: SMS, Email, Web push. SMS is effective but costly, Email is cheaper, Web push is useful for logged-in users.

Decision

Phase 1: Email + Web push. Phase 2: Add SMS for critical reminders.

Status

Accepted

Consequences

- Low initial cost with Email.
- Less reliable if customers don't check emails.
- Requires additional infra for SMS in Phase 2.